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Generative AI for RAG based AI system for Automated Telecom Customer Support

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Research Proposal

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Abstract

The telecommunications industry faces increasing challenges in managing large volumes of customer inquiries while maintaining high-quality, accurate, and timely support. Traditional customer service systems often struggle to provide context-aware responses, leading to longer resolution times and reduced customer satisfaction. This research proposes the development of a Retrieval-Augmented Generation (RAG)-based Artificial Intelligence (AI) system for automated telecom customer support. The primary objective of this study is to enhance customer service efficiency by integrating Large Language Models (LLMs) with telecom-specific knowledge retrieval mechanisms. The proposed system will utilize telecom datasets, including FAQs, troubleshooting guides, billing information, service plans, and policy documents. Data will be processed and stored in a vector database to enable efficient retrieval of relevant information. The retrieved knowledge will be combined with generative AI techniques to produce accurate, context-aware, and real-time responses to customer queries. The expected outcome is an intelligent customer support system capable of improving response accuracy, reducing query resolution time, minimizing AI hallucinations, and increasing customer satisfaction. System performance will be evaluated using metrics such as retrieval accuracy, response relevance, resolution efficiency, and user satisfaction. The study will also examine important considerations including data privacy, security, transparency, and responsible AI deployment. The findings are expected to demonstrate that RAG-based AI systems can significantly improve telecom customer support operations by delivering scalable, reliable, and cost-effective automated assistance while enhancing the overall customer experience.

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1. Background

One of the major challenges faced by telecommunication service providers is providing accurate, timely, and context-aware customer support for increasingly complex products, network services, and technical issues. Modern telecom networks generate enormous amounts of information, including 3GPP standards, operational manuals, troubleshooting guides, network logs, customer interaction records, and technical documentation. As telecom technologies continue to evolve with the adoption of 5G, cloud-native networks, Open RAN, and Software-Defined Networking (SDN), customer support engineers and traditional chatbot systems often struggle to interpret complex technical queries and deliver consistent responses (Yilma et al., 2025). Conventional rule-based customer support systems are limited in their ability to understand user intent, retrieve relevant information from large knowledge repositories, and generate accurate responses for domain-specific telecom problems. As a result, customers frequently experience delayed resolutions, inconsistent support quality, and increased dependency on human experts.

The rapid advancement of Artificial Intelligence (AI), particularly Generative AI and Large Language Models (LLMs), has created new opportunities for transforming automated customer support systems. Recent developments in Retrieval-Augmented Generation (RAG) have enabled AI systems to combine the reasoning capabilities of large language models with external knowledge retrieval mechanisms, allowing them to generate more accurate and contextually relevant responses (Lewis et al., 2020). Unlike traditional language models that rely solely on pre-trained knowledge, RAG-based systems retrieve relevant information from enterprise knowledge bases, vector databases, technical documentation, and telecom standards before generating responses. This approach significantly improves factual accuracy, reduces hallucinations, and enables AI systems to provide reliable answers for complex telecom customer queries (Analytics, D. and Sukhwal, D. (2024)).

The increasing availability of telecom knowledge repositories, customer service records, operational support systems, and technical documentation has generated vast amounts of enterprise data that can be effectively utilized to improve automated customer support. At the same time, advances in Generative AI, dense retrieval, vector databases, semantic search, and prompt

engineering have significantly enhanced the capability of AI systems to understand natural language, retrieve relevant technical information, and generate human-like responses. These technologies enable automated customer support systems to deliver more accurate, explainable, and personalized responses while reducing response time and improving customer satisfaction.

Possible future applications of Generative AI in telecom customer support are extensive, particularly in automating technical assistance, improving knowledge retrieval, and supporting customer service engineers in resolving complex telecom issues. The integration of Generative AI with Retrieval-Augmented Generation can provide intelligent retrieval of relevant technical documents, generate context-aware troubleshooting recommendations, and continuously utilize updated enterprise knowledge without requiring frequent model retraining (Bornea et al., 2025). The use of Generative AI can therefore improve response accuracy, reduce hallucinations, enhance customer satisfaction, and support scalable AI-powered customer service across modern telecommunication networks (Yilma et al., 2025). The present study will leverage the capabilities of Generative AI and Retrieval-Augmented Generation to develop an intelligent AI system for automated telecom customer support that improves retrieval effectiveness, response quality, and operational efficiency while enabling reliable and scalable deployment in enterprise telecom environments.

2. Related Work

Bornea et al. (2025) proposed Telco-oRAG, a Retrieval-Augmented Generation (RAG) framework that combines hybrid retrieval and neural routing to improve the accuracy of telecom query responses. Their study demonstrated better retrieval precision and response quality for telecom-specific knowledge. However, the research primarily focused on retrieval performance and paid limited attention to memory efficiency, scalability, and deployment in real-world telecom environments. The authors recommended further research on reducing model latency, improving resource efficiency, and developing more scalable and trustworthy AI systems for enterprise telecom applications.

Analytics, D. and Sukhwal, D. (2024) explain that Retrieval-Augmented Generation (RAG) chatbot for customer support by integrating external knowledge retrieval with Large Language Models (LLMs). The study investigated keyword extraction, prompt engineering, and document

retrieval using Mean Reciprocal Rank (MRR) and Recall@k as evaluation metrics. The proposed approach improved retrieval accuracy and response relevance; however, it mainly focused on the retrieval component rather than the overall end-to-end RAG system. The research also relied on a company-specific dataset and did not explore advanced techniques such as hybrid retrieval, reranking, or adaptive retrieval, highlighting opportunities for developing more scalable and domain-independent RAG systems.

Yuan et al. (2025) investigated the hybrid framework that integrates Knowledge Graphs (KGs) with Retrieval-Augmented Generation (RAG) to improve Large Language Models (LLMs) for telecommunications. The proposed approach significantly enhanced domain-specific question answering and knowledge grounding compared with standalone LLM and RAG models. However, the study mainly focused on technical question answering and did not evaluate broader applications such as automated telecom customer support, network management, or real-time decision-making.

Singh et al. (2025) explored the application of Artificial Intelligence (AI) in telecommunications customer support using machine learning, natural language processing, predictive analytics, chatbots, and virtual assistants. Their study showed that AI-based solutions improved troubleshooting, reduced response time, and enhanced customer satisfaction. However, the research mainly focused on conventional AI techniques and did not explore advanced Generative AI, Large Language Models (LLMs), or Retrieval-Augmented Generation (RAG) for intelligent and context-aware customer support.

Kim and Lee (2010) examined the factors influencing customer loyalty in the mobile telecommunications industry using survey data from 469 customers and multiple regression analysis. Their findings showed that corporate image, service quality, brand awareness, and service price significantly affect customer loyalty, with corporate image having the strongest influence. However, the study focused on traditional marketing factors and did not examine the impact of AI-driven customer support, predictive analytics, or personalized digital services on customer loyalty.

Huang et al. (2015) investigated the use of big data analytics for customer churn prediction in the telecommunications industry by analysing large-scale Business Support Systems (BSS) and Operations Support Systems (OSS) datasets. Their findings showed that big data techniques significantly improved churn prediction accuracy by capturing customer behavioural and usage

patterns. However, the study primarily focused on traditional big data analytics and did not explore advanced AI approaches such as deep learning, graph-based learning, or Generative AI for intelligent telecom applications.

Ahmad et al. (2019) exmined that machine learning techniques on the Apache Spark big data platform to predict customer churn in the telecommunications industry. Their study analysed large-scale BSS and OSS datasets and found that the XGBoost algorithm achieved the highest performance with an AUC of approximately 93%, while Social Network Analysis (SNA) further improved prediction accuracy. However, the research focused on customer churn prediction and did not investigate Generative AI, Large Language Models (LLMs), or Retrieval-Augmented Generation (RAG) for intelligent telecom customer support.

Jadhav et al.(2011) have studied how data mining techniques for customer churn prediction in the telecommunications industry. Their study applied classification and clustering methods to analyse telecom customer data and found that effective data preprocessing, including data cleaning, feature selection, and handling missing values, significantly improved churn prediction accuracy. However, the research relied on traditional data mining techniques and did not explore advanced Artificial Intelligence, deep learning, or Retrieval-Augmented Generation (RAG) approaches for intelligent telecom applications.

Amatare and Ojo (2020) have studied used of data mining techniques for customer churn prediction in the telecommunications industry. Their study applied classification and clustering algorithms on structured telecom customer datasets and found that feature selection and handling missing values significantly improved prediction performance. However, the research focused on traditional data mining methods and did not explore advanced Artificial Intelligence, deep learning, ensemble learning, or Retrieval-Augmented Generation (RAG) techniques for intelligent telecom customer support.

Fujo et al. (2022) investigated the Artificial Intelligence and deep learning techniques for customer churn prediction using the IBM Telco Customer Churn dataset (7,043 customers) and the Cell2Cell dataset (51,047 customers). Their Deep Backpropagation Artificial Neural Network (Deep-BP-ANN) outperformed XGBoost, Logistic Regression, Naïve Bayes, and KNN based on metrics such as Accuracy, Precision, Recall, F1-Score, and AUC. However, the study focused on customer

churn prediction and did not investigate Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or conversational AI for automated telecom customer support.

Patel and Trivedi (2020) has investigated the the use of Artificial Intelligence (AI), Machine Learning (ML), Natural Language Processing (NLP), and AI chatbots to improve customer loyalty and support services. Their study analysed customer interaction data using algorithms such as Logistic Regression, KNN, SVC, Decision Tree, Random Forest, AdaBoost, Extra Trees, and Bagging Classifier, evaluated through accuracy, F1-score, confusion matrix, and ROC curve. The results showed that AI-powered chatbots improved customer engagement; however, the study did not incorporate Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or knowledge-grounded response generation, highlighting the need for more intelligent and context-aware customer support systems.

Sabbeh et al. (2018) conducted a comparative study of machine learning algorithms for customer retention and churn prediction using a telecommunications dataset containing customer billing and interaction records. The study evaluated algorithms including Decision Tree, Logistic Regression, Support Vector Machine (SVM), Random Forest, AdaBoost, and Stochastic Gradient Boosting using 10-fold cross-validation and classification accuracy. Although the study identified effective churn prediction models, it mainly relied on traditional machine learning techniques and did not explore Generative AI, Large Language Models (LLMs), or Retrieval-Augmented Generation (RAG) for intelligent telecom customer support.

Alkhayrat et al. (2020) investigated the application of Artificial Intelligence (AI) and Deep Learning (DL) techniques for telecom customer segmentation using the SyriaTel Telecom dataset containing 100,000 customer records with 220 features. The study compared Autoencoder Neural Networks with Principal Component Analysis (PCA) for dimensionality reduction, followed by K-Means clustering, and found that Autoencoders achieved better clustering performance based on MSE, Silhouette Coefficient, DBI, and SSE. However, the research focused on customer segmentation and did not explore Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or conversational AI for intelligent telecom customer support.

Ullah et al. (2019) investigated the use of Artificial Intelligence (AI) and Machine Learning (ML) techniques for customer churn prediction using a real-world telecom dataset containing customer usage, call records, revenue, and behavioural data. The study compared algorithms including Random Forest, J48 Decision Tree, SVM, Logistic Regression, Naïve Bayes, KNN, and K-Means Clustering, and found that Random Forest and J48 achieved the highest performance based on Accuracy, Precision, Recall, F-measure, ROC Area, and Cross-Validation. However, the research focused on churn prediction and customer retention, without exploring Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), or conversational AI for automated telecom customer support.

Table 1: Existing Research Summarization

Ref	Used Dataset	Used Algorithms	Used Evaluation Metrics	Seen Research Gaps
[1]	Telecom Standards(3GPP specification, telecom technical)	RAG, LLMs, Dense Retrieval, Vector Databases embeddings and dialouge history tracking.	Answer Accuracy, Context Relevance, Retrieval Precision, Hallucination Rate	Focuses on telecom standards rather than real-world customer support interactions and customer experience.
[2]	China telecom customer services dataset	NLP, Machine learning based Automation, AI, Conversational AI, Intellegent customer service systems	Customer Satisfaction (CSAT),Service effeciency, Response time, operational performance	Lack of evaluation of LLM for complex customer support and telecom assistance.
[3]	Custom customer-support dataset	RAG, LLM, Extraction, prompt, OpenSearch(Elasticsearch), DenseRetrieval,Top-5 Document Retrieval	(MRR), Recall@n, Retrieval Accuracy, Keyword Generation Performance, Prompt Comparison	focuses primarily on retrieval evaluation rather than complete end-to-end response generation. The evaluation is limited to a company-specific customer-support dataset.

[4]	Telecommunications domain documents dataset	(LLMs), (RAG), Knowledge Graphs (KGs), Dense Retrieval, Vector Embeddings, Hybrid KG-RAG Framework	Answer Accuracy, Response Relevance, Retrieval Effectiveness, Knowledge Grounding/Factual Correctness (comparison of LLM-only, RAG-only, and KG-RAG approaches)	Limited evaluation of real-world customer support interactions, customer experience management, network operations, predictive maintenance, and real-time decision-support applications.
[5]	Telecom customer interaction logs, support tickets, and network operational dataset.	NLP, ML, Predictive Analytics, Chatbots, Virtual Assistants, Sentiment Analysis.	Resolution Accuracy, Response Time, Customer Satisfaction, First Contact Resolution, Service Efficiency.	Limited investigation of LLMs, Knowledge Graphs, RAG, and advanced context-aware conversational support for complex troubleshooting tasks.
[6]	Web-based survey dataset	Survey Research, Multiple Regression Analysis, Correlation Analysis.	Customer Loyalty, Corporate Image, Brand Awareness, Service Price, Service Quality, Customer Support Service, Statistical Significance (p-values).	Limited investigation of digital transformation, AI-enabled customer support, personalized customer experiences, social media influence, and predictive customer retention strategies
[7]	Large-scale telecom datasets from BSS and OSS	Big Data Analytics,ML-based classification models, feature engineering on BSS/OSS data, distributed data processing frameworks.	Precision (e.g., top-k churn prediction precision), predictive accuracy, retention rate improvement, business value metrics.	Focuses on traditional big-data-driven churn prediction using structured telecom data. Limited exploration of deep learning, graph-based models, LLMs, and AI-driven personalized

				customer retention systems.
[8]	Large-scale telecom customer datasets	ML models (including tree-based classifiers such as Random Forest and XGBoost), feature engineering techniques, Spark-based distributed data processing, and (SNA)	AUC, Accuracy, Precision, Recall, F1-score, and churn prediction performance improvement after feature enhancement and SNA integration.	Low diversity in deep learning approaches, real-time streaming analytics, graph neural networks, and modern AI techniques such as LLMs
[9]	Structured telecom CRM dataset	Data Mining techniques including classification algorithms, along with data preprocessing and feature selection techniques.	Accuracy, Precision, Recall, F1-score, and churn classification performance	Limited exploration of big data frameworks, deep learning models, graph-based learning, real-time analytics.
[10]	Telecom customer dataset using customer demographics	Decisiuon tree, ML feature selection, clustering methods	Churn prediction models, Overall classification performance, F1-Scores, Recall	Reliance on datasets, limited model complexity, lack of real-time analysis, absence of deep learning methods
[11]	IBM Telco Customer Churn Dataset (7,043 customers) and Cell2Cell Dataset (51,047 customers)	Deep Backpropagation Artificial Neural Network (Deep-BP-ANN), XGBoost, Logistic Regression (LR), Naïve Bayes (NB), K-Nearest Neighbour (KNN),	Accuracy (ACC), Precision, Recall, F1-Score, Area Under the Curve (AUC), Hold-out Validation, 10-Fold Cross Validation	The study focuses on customer churn prediction rather than automated telecom customer support. It does not incorporate Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), real-time knowledge retrieval, explainable AI, personalized customer assistance, or conversational AI.

[12]	Customer loyalty dataset	Logistic Regression, (KNN), Support Vector Classifier (SVC), Decision Tree, Random Forest, AdaBoost,	Training Accuracy, Testing Accuracy, F1-Score, Confusion Matrix, ROC Curve	The study concludes that existing AI chatbots cannot effectively handle complex or situational customer queries and often appear unnatural and insensitive. It recommends improving chatbot personalization, human-like interactions (anthropomorphism), trust, and privacy.
[13]	Public Telecom Customer Churn Dataset containing 3,333 customer records with 17 explanatory features	Logistic Regression (LR), Decision Tree (CART), Support Vector Machine (SVM), Naïve Bayes (NB), K-Nearest Neighbour (KNN), Random Forest (RF), AdaBoost, S	10-Fold Cross Validation, Classification Accuracy, Minimum Accuracy, Maximum Accuracy. Models were trained using 60% training and 40% testing data.	The authors concluded that Random Forest and AdaBoost achieved the highest prediction accuracy (~96%). They recommended extending the work by developing hybrid models and deep learning models, using additional performance evaluation metrics, considering execution time as a performance indicator, and validating the models on different datasets from various domains.
[14]	SyriaTel Telecom customer dataset	Principal Component Analysis (PCA), Autoencoder Neural Network (Deep Learning).	Mean Squared Error (MSE), Silhouette Coefficient, Davies–Bouldin Index.	The study focused on customer segmentation and did not address AI-driven customersupport, Retrieval-Augmented Generation (RAG), LargeLanguage

				Models(LLMs), or conversational AI for telecom customer.
[15]	Real-world telecom customer dataset	Random Forest (RF), J48 Decision Tree, Naïve Bayes (NB), Support Vector Machine (SVM), Logistic Regression (LR), K-Nearest Neighbour (KNN)	Accuracy, Precision, Recall, F-Measure (F1-Score), ROC Area, and 10-Fold Cross-Validation.	The study mainly focused on customer churn prediction and retention strategies using machine learning techniques. It did not address intelligent customer support

3. Research Questions

This research investigates the application of Retrieval-Augmented Generation (RAG) integrated with Large Language Models (LLMs) to improve customer support services in the telecommunications sector. The study seeks to evaluate how domain-specific knowledge retrieval can enhance response quality, reduce hallucinations, and improve operational efficiency. The following research questions guide this investigation.

RQ1: In what ways can Large Language Models be combined with Retrieval-Augmented Generation and telecom knowledge repositories to provide accurate and context-aware responses for customer support?

RQ2: How does the proposed RAG-enabled support framework perform compared with conventional rule-based chatbot systems when evaluated using response accuracy, retrieval speed, issue resolution, and customer satisfaction?

RQ3: Which retrieval strategy—including hybrid retrieval, dense vector search, knowledge graph integration, or reranking—provides the best balance between retrieval accuracy, hallucination reduction, and inference latency within telecom support environments?

RQ4: What ethical, privacy, security, and governance factors should be addressed before deploying Generative AI-powered decision support systems in telecommunications?

4. Aim and Objectives

The overall aim of this research is to design, develop, and evaluate a Generative Artificial Intelligence system based on Retrieval-Augmented Generation (RAG) for automated telecom customer support. The proposed system aims to combine the reasoning capabilities of LLMs with telecom-specific knowledge retrieval techniques to provide accurate, context-aware, and reliable responses to customer queries. By integrating enterprise knowledge bases, semantic search, vector databases, and modern retrieval mechanisms, the study seeks to reduce AI hallucinations, improve response quality, and enhance customer satisfaction while minimizing response latency. Furthermore, the research intends to investigate the effectiveness of RAG-based architectures compared with conventional rule-based and standalone LLM chatbot systems operating in telecommunications customer service environments. The specific objectives of the study are as follows:

- To study and analyze the existing approaches for automated telecom customer support.
- To design Retrieval-Augmented Generation (RAG) framework with telecom-specific knowledge.
- To integrate Large Language Models with telecom-specific RAG for automated telecom customer support.
- To evaluate the effectiveness of semantic retrieval techniques, vector databases, and embedding models in improving response relevance and factual accuracy.
- To compare the proposed automated telecom customer support framework with conventional chatbot systems using evaluation metrics such as, Retrieval Precision, Faithfulness, Response Relevance, Hallucination Rate, Query Resolution Time, and Customer Satisfaction.

5. Significance of the Study

The significance of this study lies in addressing one of the major challenges faced by the telecommunications industry, namely providing accurate, timely, and context-aware customer support while managing an increasing volume of customer queries. As telecom networks continue to evolve with technologies such as 5G, cloud-native architectures, and Software-Defined Networking (SDN), customer service systems must process increasingly complex technical information. Traditional rule-based chatbots and standalone Large Language Models (LLMs) often struggle to generate reliable and domain-specific responses, leading to longer resolution times, inconsistent support quality, and reduced customer satisfaction (Yilma et al., 2025). This study aims to improve automated telecom customer support by developing a Retrieval-Augmented Generation (RAG)-based Artificial Intelligence system that combines semantic retrieval with Generative AI to deliver accurate, context-aware, and knowledge-grounded responses. The proposed framework is expected to reduce AI hallucinations, improve retrieval precision, enhance response relevance, and increase customer satisfaction (Analytics, D. and Sukhwal, D., 2024. Furthermore, the integration of Retrieval-Augmented Generation with Large Language Models has significant potential to transform enterprise customer support by enabling AI systems to retrieve verified telecom knowledge before generating responses. This approach improves factual correctness, transparency, and explainability while reducing operational costs and improving service efficiency (Bornea et al., 2025). In addition, the study contributes to the responsible adoption of Generative AI by examining interpretability, fairness, data privacy, and ethical AI deployment within enterprise environments. As highlighted by Yuan et al. (2025), combining advanced retrieval mechanisms with domain-specific knowledge significantly enhances the reliability of AI-generated responses. Therefore, the findings of this research are expected to support the broader adoption of RAG-based AI systems, enabling telecommunication providers to deliver scalable, trustworthy, and intelligent customer support services.

6 Scope of the Study

The particular focus of the current research is the design and development of a Generative Artificial Intelligence system based on Retrieval-Augmented Generation (RAG) for automated telecom customer support. The study will involve developing an intelligent AI model that integrates Large Language Models (LLMs) with telecom-specific knowledge retrieval mechanisms to provide accurate, context-aware, and knowledge-grounded responses to customer queries. The research will make use of publicly available telecom datasets such as the TelecomRAG dataset, 3GPP telecom standards, telecom customer support FAQs, telecom troubleshooting manuals, customer service knowledge base documents, and other publicly available telecom datasets from Kaggle, Hugging Face, and GitHub repositories. These datasets will be utilized to construct a telecom knowledge base, generate vector embeddings, and establish an efficient retrieval mechanism capable of supporting accurate response generation for customer support applications.

The research is limited to the development and evaluation of a Retrieval-Augmented Generation-based customer support system within the telecommunications domain by intentionally restricting the scope of the study. Other enterprise applications of Generative AI, such as network fault prediction, customer churn prediction, predictive maintenance, fraud detection, network optimization, or telecom resource management, will not be included in the research because they require different datasets, methodologies, and evaluation frameworks (Yilma et al., 2025). Furthermore, in as much as the research will be based on the application of Generative AI for intelligent customer support, it will not address commercial deployment, live CRM integration, billing transactions, voice assistants, or multilingual speech processing, but rather the development of a knowledge-grounded conversational AI system that can be achieved through Retrieval-Augmented Generation. The ethical and interpretability concerns of Artificial Intelligence, including transparency, explainability, fairness, bias mitigation, and data privacy, will also be examined but limited to the deployment of RAG-based customer support systems within enterprise telecommunications environments.

7. Research Methodology

This research will use a quantitative experimental research design to develop and evaluate a Retrieval-Augmented Generation (RAG) based Artificial Intelligence system for automated telecom customer support. The main objective is to investigate whether combining Large Language Models (LLMs) with telecom-specific knowledge retrieval can provide more accurate, relevant, and context-aware responses than traditional rule-based chatbot systems. The complete automated telecom customer support framework is shown in Figure 1. The complete research methodology for this framework is explained below.

7.1.Data Collection:

This study will use publicly available telecom datasets and technical documents collected from reliable academic and industrial sources. The primary dataset will be TelecomRAG (Yilma et al., 2025), which contains telecom standards, operational manuals, 3GPP specifications, and other technical documents suitable for Retrieval-Augmented Generation research. Additional telecom query datasets developed by Bornea et al. (2025) and customer support datasets evaluated by Sukhwal and Analytics (2024) will also be included to improve the diversity of customer queries and domain knowledge.

DataSet Details:

TelecomRAG (Yilma et al.): It will offers RAG Based framework which using the 3GPP release 16 and release 18 standards for telecom questions answering, which will be utilized in RAG research.

RAG Customer Support Dataset (Sukhwal & Analytics, 2024): This Dataset came with evaluation a RAG based customer support document collection using in RAG evaluation, it can be used in RAG based chatbots performance.

3GPP Telecom Standards: Its availble at keggle and one of the most important knoledge source for Telecom RAG systems and also using in extensively in Telecom RAG and telco oRAG,will be used

7.2 Data Pre-processing:

Before the datasets are used for model development, they will be cleaned and prepared through several preprocessing steps, including removing duplicate records, standardizing the text, splitting large documents into smaller sections, and extracting useful metadata. Excel will be used to organise and validate the collected data, while Python will be used for text preprocessing and embedding generation. The final output will be a structured telecom knowledge base that supports efficient semantic retrieval and accurate response generation.

7.3. Vector Database Deployment:

After preprocessing, the generated document embeddings will be stored in a vector database such as FAISS or ChromaDB. The vector database will organize telecom knowledge including billing information, troubleshooting guides, and technical documentation to enable fast and accurate semantic search. This process will improve the retrieval of relevant documents based on customer queries.

7.4. RAG Model Design and Development:

The proposed customer support system will be developed by combining a Large Language Model (LLM) with a Retrieval-Augmented Generation (RAG) framework. When a customer asks a question, the system will first search the vector database to retrieve relevant telecom information before generating a response. This approach is expected to improve response accuracy, provide better context, and reduce AI hallucinations.

7.5. Model Evaluation and Performance Analysis:

The performance of the proposed system will be evaluated using measures such as Answer Accuracy, Retrieval Precision, Recall@k, Mean Reciprocal Rank (MRR), Response Relevance, Hallucination Rate, and Query Resolution Time. The experimental results will be analysed using SPSS and compared with traditional rule-based chatbots and standalone LLM models. This evaluation will help determine whether the proposed framework provides more accurate and reliable customer support.

7.6. Ethical Considerations and Final Analysis:

The final stage of the research will focus on evaluating the ethical aspects of the proposed system, including transparency, fairness, privacy, and responsible AI practices. The responses generated by the model will be reviewed to ensure they are supported by reliable telecom knowledge and minimise hallucinations. Finally, the proposed framework will be compared with existing customer support systems to assess its overall performance and practical applicability. Only publicly available telecom datasets and technical documentation will be used during the study. No confidential customer information or personally identifiable data will be collected or processed. To improve interpretability, every generated response will be linked to the retrieved telecom documents that were used as supporting evidence. This will allow researchers and users to understand how the response was generated and verify whether it is based on reliable information.

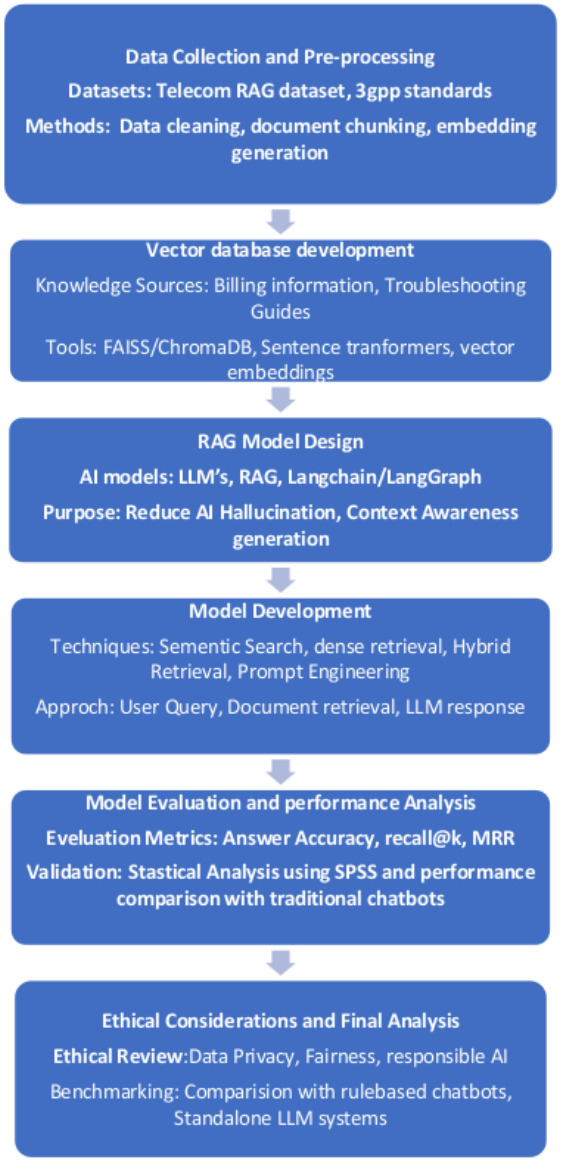


Figure 1: Framework Flowchart

8. Requirements and Resources

Software:

Microsoft Excel:

Microsoft Excel will be employed during the initial stages of the research for organising, inspecting, and preparing the collected datasets. It will assist in identifying missing or inconsistent values, performing preliminary data transformations, and generating descriptive summaries and

visual representations. These activities will support data understanding before the datasets are processed within the proposed RAG framework.

IBM SPSS Statistics: IBM SPSS Statistics will be used to perform quantitative analysis and evaluate the effectiveness of the proposed framework. The software will support descriptive and inferential statistical techniques, including regression analysis, hypothesis testing, independent sample t-tests, ANOVA, and correlation analysis where appropriate. Statistical findings will be used to compare the proposed RAG-enabled customer support system with conventional chatbot approaches across evaluation measures such as response accuracy, retrieval efficiency, customer satisfaction, and overall system performance.

Python: Python will be used to implement the Retrieval-Augmented Generation framework. It will support data preprocessing, embedding generation, semantic retrieval, Large Language Model integration, evaluation, and visualization using libraries such as LangChain, LlamaIndex, Hugging Face Transformers, Pandas, NumPy, and Scikit-learn.

Hardware:

Computing Resources: The research will involve the computer with sufficient processing power to develop and evaluate the RAG framework. A workstation equipped with an Intel Core i7 or AMD Ryzen 7 processor, at least 16 GB RAM, SSD storage, and access to an NVIDIA GPU will provide adequate computational resources for embedding generation and model evaluation, also requires the cloud platforms like AWS(Sagemaker AI), Azure(AI foundary), Google cloud(Vertex AI) used.

Table 2: Software and hardware requirements for the study

Software	Hardware	Datasets
Microsoft Excel, IBM SPSS Statistics, Python, LangChain, LlamaIndex, Hugging Face Transformers, Pandas, NumPy, Scikit-learn, FAISS (or ChromaDB), Matplotlib	Intel Core i7/AMD Ryzen 7 or Xeon processor, 16–32 GB RAM, SSD storage (minimum 256 GB), NVIDIA GPU (recommended), AWS SageMaker, Azure AI Foundry, or Google Vertex AI	TelecomRAG (Yilma et al., 2025), Telco-ORAG (Bornea et al., 2025), RAG Customer Support Dataset (Sukhwal & Analytics, 2024), Telecom FAQs, 3GPP standards, troubleshooting manuals, customer service documentation

Supports data preparation, statistical evaluation, implementation of the RAG pipeline, semantic retrieval, visualisation, and performance assessment	Provides the computational capability required for embedding generation, vector indexing, LLM inference, and experimental evaluation	Supplies domain-specific knowledge for retrieval, response generation, benchmarking, and evaluation of the proposed telecom customer support framework

9. Research Plan

Phase 1: Literature Review and Data Collection(2 Weeks)

The research will begin with a systematic examination of academic publications, industry reports, technical standards, and recent studies related to Generative Artificial Intelligence, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), semantic information retrieval, vector databases, and intelligent telecom customer support systems. This review aims to establish the current state of research, identify unresolved challenges, and determine opportunities for improving AI-powered customer support within the telecommunications industry.

Relevant datasets and domain-specific documents will then be collected to support the development of the proposed framework. These resources include TelecomRAG (Yilma et al., 2025), Telco-ORAG (Bornea et al., 2025), the RAG Customer Support Dataset (Sukhwal & Analytics, 2024), telecom Frequently Asked Questions (FAQs), troubleshooting guides, billing documentation, customer service manuals, and 3GPP technical standards. Together, these materials will provide the knowledge repository required for implementing the Retrieval-Augmented Generation system.

Phase 2: Data Preprocessing and Model Development(3 Weeks)

In this step, the data will be preprocessed and arranged in excel will be cleaned and prepared for use in the proposed system. This process will include removing duplicate records, correcting formatting issues, normalising the text, and dividing large documents into smaller sections to improve retrieval performance.After preprocessing, document embeddings will be generated and stored in a vector database such as FAISS or ChromaDB. The Retrieval-Augmented Generation framework will then be developed by integrating the vector database with a Large Language Model

Phase 3: Model Evaluation and Statistical Analysis (3 weeks):

The completed RAG framework will be evaluated using quantitative performance measures, including response accuracy, precision, recall, F1-score, retrieval latency, and customer query resolution effectiveness. Statistical analysis will be performed using IBM SPSS Statistics to determine whether the proposed approach demonstrates measurable improvements over conventional rule-based customer support systems.

Descriptive statistics, correlation analysis, independent sample t-tests, ANOVA, and regression analysis will be applied where appropriate to examine relationships between evaluation variables and validate the research findings. The statistical outcomes will support conclusions regarding the reliability, effectiveness, and practical applicability of the proposed framework.

Phase 4: Final Review, Interpretability, and Analysis (3 Weeks)

The final phase will focus on interpreting the experimental findings and assessing the overall performance of the proposed RAG-based customer support framework. The generated responses will be analysed for contextual relevance, factual consistency, retrieval quality, and the occurrence of hallucinations. The strengths and limitations of the proposed solution will also be discussed in relation to existing telecom customer support methods.

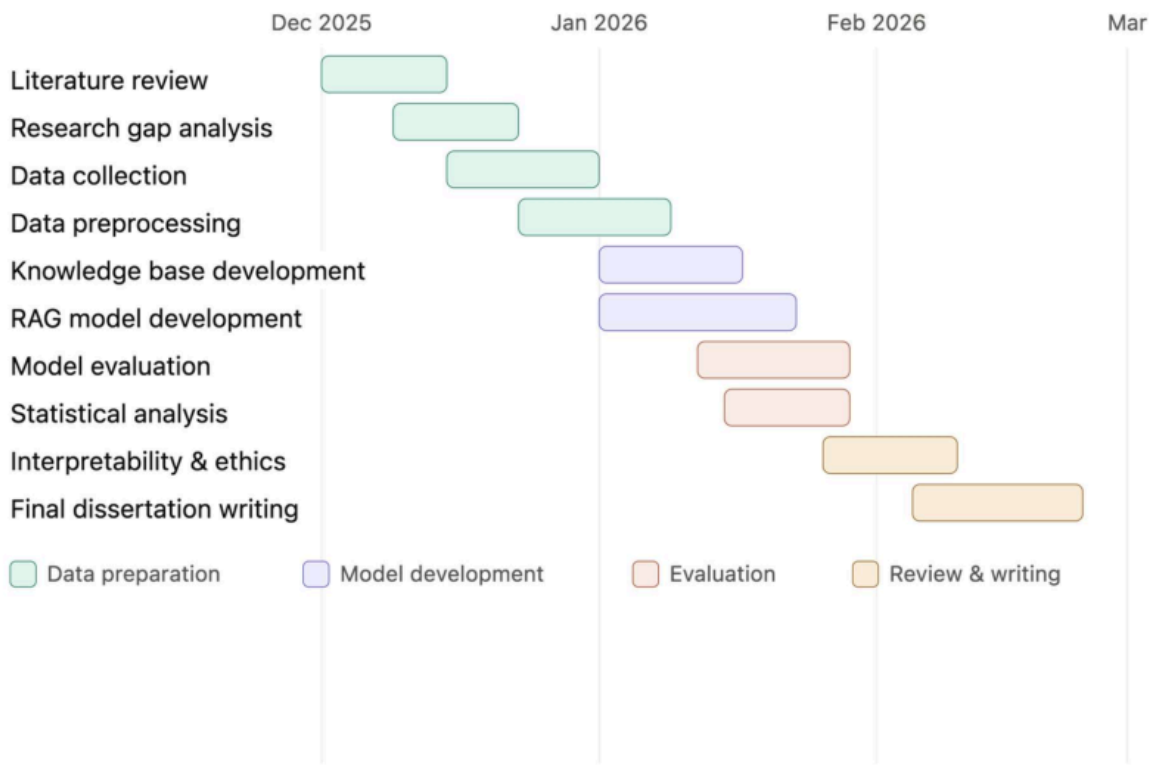


Figure 2: Proposed Research Plan

10. Risks and Contingency Plans

Several challenges may arise during the course of this research. These include limited availability of high-quality telecom datasets, incomplete or inconsistent technical documentation, retrieval errors, AI hallucinations, and computational resource limitations. Variations in telecom terminology across different datasets may also affect retrieval accuracy and the overall performance of the proposed system.

To minimise these risks, multiple publicly available telecom datasets will be combined to improve the quality and coverage of the knowledge base. Advanced retrieval techniques, including hybrid retrieval, reranking methods, and prompt optimisation, will be used to improve retrieval accuracy and reduce hallucinations. Cross-validation and repeated experimental testing will be carried out to ensure that the results are reliable and consistent. If local hardware resources become insufficient, cloud-based platforms such as Google Colab, AWS SageMaker, Azure AI Foundry, or Google Cloud Vertex AI will be used to complete the experiments successfully.

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